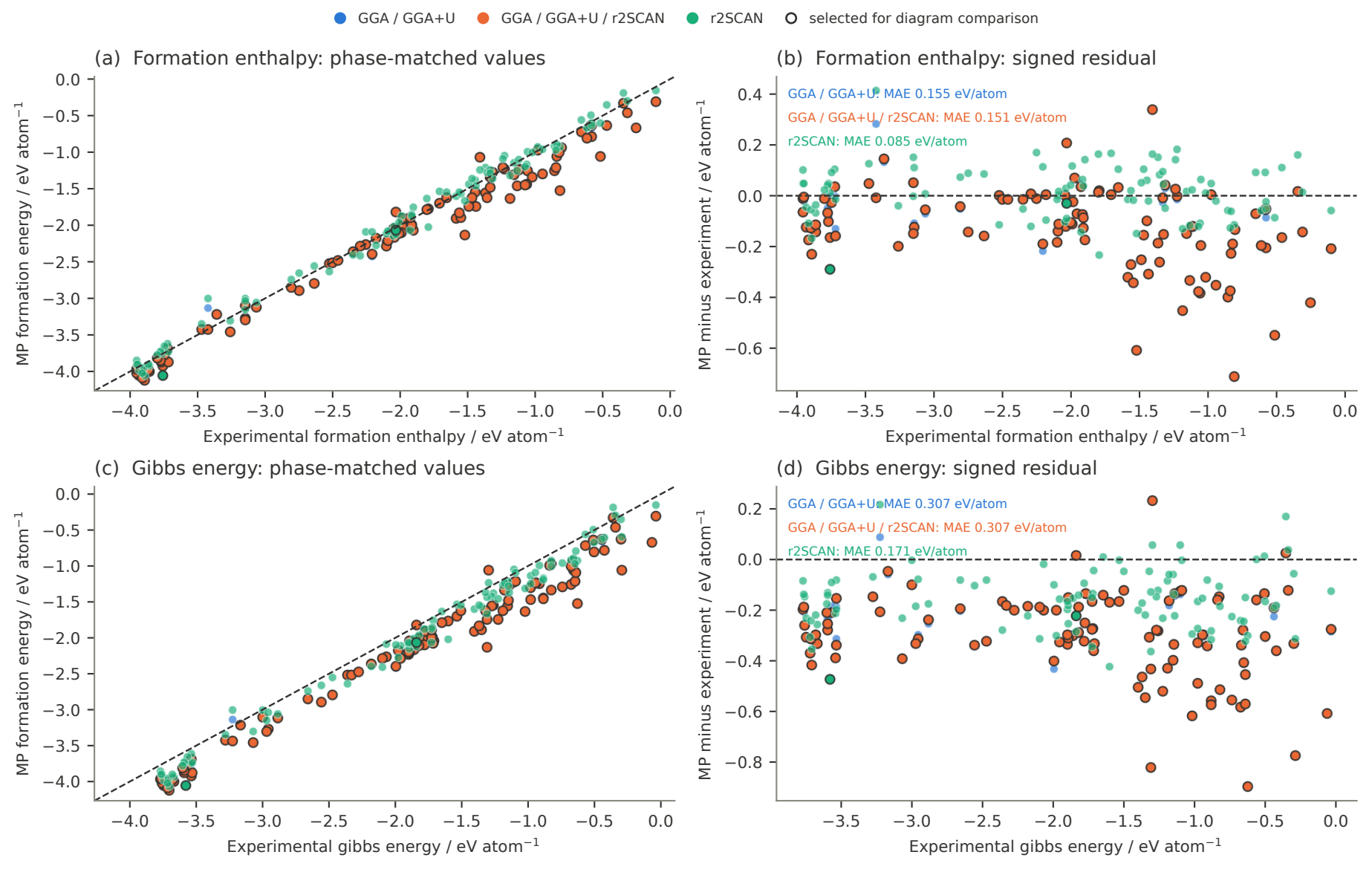

GGA / GGA+U
GGA / GGA+U / r2SCAN
r2SCAN
selected for diagram comparison
(a) Formation enthalpy: phase-matched values
MP formation energy / eV atom$^{-1}$
Experimental formation enthalpy / eV atom$^{-1}$
(b) Formation enthalpy: signed residual
GGA / GGA+U: MAE 0.155 eV/atom
GGA / GGA+U / r2SCAN: MAE 0.151 eV/atom
r2SCAN: MAE 0.085 eV/atom
MP minus experiment / eV atom$^{-1}$
Experimental formation enthalpy / eV atom$^{-1}$
(c) Gibbs energy: phase-matched values
MP formation energy / eV atom$^{-1}$
Experimental gibbs energy / eV atom$^{-1}$
(d) Gibbs energy: signed residual
GGA / GGA+U: MAE 0.307 eV/atom
GGA / GGA+U / r2SCAN: MAE 0.307 eV/atom
r2SCAN: MAE 0.171 eV/atom
MP minus experiment / eV atom$^{-1}$
Experimental gibbs energy / eV atom$^{-1}$